3.4 Vectors and Dot Products

Name _____ Period ____ Date _____

Pre-Calculus 3.4 (30-minute plan) · Student sheet

I can…

- find the dot product of two vectors and use its properties.
- find the angle between two vectors.
- decide whether two vectors are orthogonal.

Vocabulary · fill in the missing word

Dot product — $u \cdot v = u_1v_1 + u_2v_2$ (a number, not a _____)

Angle between vectors — $\cos\theta =$ _____, $0 \le \theta \le \pi$

Orthogonal — _____: $u \cdot v = 0$

Part 1 · Notes and examples (fill in each note, then work the example)

Dot product: $u \cdot v =$ _____. The answer is a _____.

Example 1

Find each dot product.

a. $\langle 4, 5\rangle \cdot \langle 2, -3\rangle$ b. $\langle 0, 3\rangle \cdot \langle 4, -2\rangle$

Properties: $u \cdot u =$ _____

Angle between vectors: Angle between u and v: $\cos\theta =$ _____

Example 2

Find the angle between $u = \langle 4, 3\rangle$ and $v = \langle 3, 5\rangle$.

Orthogonal vectors: u and v are **orthogonal** when _____.

Example 3

Are $u = \langle 2, -3\rangle$ and $v = \langle 6, 4\rangle$ orthogonal?

Part 2 · Partner practice

Partner practice

Find k so that $\langle 2, k \rangle$ and $\langle 3, -6 \rangle$ are orthogonal.